

CSCE 20004

Programming Foundations I

Programming Project 1 Report

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Academic Integrity Statement

I pledge that I have neither given nor received unauthorized help on this programming assignment. I have written this report myself in my own words.

1 Problem Statement

I was hired as head chef at a restaurant. I need to create a program that allows me to scale American recipes to metric measurements. First, the program should ask how many the original recipe makes. The program should take the amount of servings I need to make. Then, it should ask me the pricing of the ingredients in USD. After, we should calculate the cost of each serving, first in USD, then in EUR. Finally, we can tell how many times we must multiply each ingredient in order to properly scale our recipe.

2 Design

I/O is handled via `std::cin` & `std::cout`. While not required, I still handled invalid input via checking `std::cin` error flag for its `failbit`: `validateInput()`. This function wraps `std::cin` & `std::cout`. This works because C++ sets the failbit whenever the input type doesn't fit the destination type (e.g, trying to store a `std::string` into a `float` via `std::cin`).

2.1 Pros:

- Code is modular and reusable
- Input checking is robust enough for the purposes of the program

2.2 Cons:

- The program ends immediately on a calculation, forcing the user to rerun it.
- It will accept unreasonably large inputs as I don't check an upper bound.

3 Implementation

I did not use any code to start my program. I got the information I needed from instructions, as well the sample output screenshots given. My development timeline was as follows:

1. Recreate outputs with `std::cout`.
2. Implement arithmetic based on instructions provided
3. Refactor code once logic works:
 - (a) Declare variables and replace hard-coded test values
 - (b) Create functions out of reused code (`validateInput()`)

4 Testing

Testing consisted of throwing in edge cases which could cause the program to fail.

Test results

Example of a good run:

```
Welcome to the Recipe Scaling Program!

Congratulations on getting hired as Chef!
Let's work on scaling your American recipes to metric measurements and ingredient costs for
↪ different serving sizes.

Please enter the following details:
How many servings does your recipe make? 23
How many servings do you want to scale it to? 68
What is the cost of your ingredients in USD? $142

Cost per serving in USD:      $6.17
Cost of ingredients in EUR:   €120.70
Cost per serving in EUR:     €5.25
To scale your recipe from 23.00 to 68.00, you will need to multiply each ingredient by 2.96
```

Example of a run which invokes the error handling:

```
Welcome to the Recipe Scaling Program!

Congratulations on getting hired as Chef!
Let's work on scaling your American recipes to metric measurements and ingredient costs for
↪ different serving sizes.

Please enter the following details:
How many servings does your recipe make? cat
[!] Please use numbers.
How many servings does your recipe make? 42
How many servings do you want to scale it to? 0
[!] Input must be greater than zero.
How many servings do you want to scale it to? 23
What is the cost of your ingredients in USD? $-643
[!] Input must be greater than zero.
What is the cost of your ingredients in USD? $120

Cost per serving in USD:      $2.86
Cost of ingredients in EUR:   €102.00
Cost per serving in EUR:     €2.43
```

5 Conclusions

The overall result of the assignment was a success as I was able to complete all requirements. I already have *some* knowledge of C++, thus this was a simple project. Overall, the project took approximately two hours to complete, including the report, initially. I then added input validation and further professionalized the code.